

Ligjerata 2ë

Lëvizja vijepërkulët

Nxitimi tangjencial

$$\varphi = \frac{l}{r}$$

$$\Delta\varphi = \frac{\Delta s}{r}$$

$$\Delta s = r \cdot \Delta\varphi$$

$$v = \frac{\Delta s}{\Delta t} \rightarrow \Delta s = v \Delta t$$

$$v \cdot \Delta t = r \cdot \Delta\varphi \quad / : \Delta t$$

$$v = r \frac{\Delta\varphi}{\Delta t} \quad ; \quad \langle \omega \rangle = \frac{\Delta\varphi}{\Delta t} = \omega$$

$$v_t = r \cdot \omega$$

$$r = \text{konst.}$$

$$a_t = \frac{dv_t}{dt} = \frac{d}{dt}(r \cdot \omega)$$

$$a_t = \omega \frac{dr}{dt} + r \frac{d\omega}{dt}$$

$$a_t = r \frac{d\omega}{dt} \quad ; \quad \frac{d\omega}{dt} = \alpha$$

$$a_t = r \alpha$$

